Test Series Code

05

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 08/02/2026

ANSWER KEY

1.	(2)	46.	(4)	91.	(2)	136.	(1)
2.	(1)	47.	(4)	92.	(1)	137.	(1)
3.	(4)	48.	(4)	93.	(1)	138.	(1)
4.	(1)	49.	(3)	94.	(2)	139.	(3)
5.	(2)	50.	(2)	95.	(1)	140.	(1)
6.	(4)	51.	(1)	96.	(4)	141.	(3)
7.	(4)	52.	(2)	97.	(2)	142.	(3)
8.	(4)	53.	(4)	98.	(1)	143.	(2)
9.	(1)	54.	(2)	99.	(3)	144.	(3)
10.	(1)	55.	(4)	100.	(3)	145.	(3)
11.	(4)	56.	(2)	101.	(1)	146.	(2)
12.	(2)	57.	(1)	102.	(4)	147.	(3)
13.	(2)	58.	(3)	103.	(1)	148.	(3)
14.	(1)	59.	(3)	104.	(2)	149.	(2)
15.	(2)	60.	(3)	105.	(4)	150.	(1)
16.	(2)	61.	(2)	106.	(1)	151.	(3)
17.	(1)	62.	(4)	107.	(2)	152.	(2)
18.	(1)	63.	(1)	108.	(3)	153.	(1)
19.	(1)	64.	(3)	109.	(2)	154.	(2)
20.	(2)	65.	(2)	110.	(2)	155.	(4)
21.	(2)	66.	(1)	111.	(1)	156.	(3)
22.	(4)	67.	(3)	112.	(2)	157.	(3)
23.	(3)	68.	(3)	113.	(4)	158.	(4)
24.	(4)	69.	(3)	114.	(3)	159.	(3)
25.	(4)	70.	(1)	115.	(1)	160.	(2)
26.	(4)	71.	(4)	116.	(4)	161.	(3)
27.	(1)	72.	(1)	117.	(2)	162.	(2)
28.	(2)	73.	(2)	118.	(3)	163.	(2)
29.	(1)	74.	(2)	119.	(4)	164.	(1)
30.	(4)	75.	(4)	120.	(3)	165.	(2)
31.	(2)	76.	(3)	121.	(1)	166.	(3)
32.	(3)	77.	(4)	122.	(3)	167.	(4)
33.	(2)	78.	(3)	123.	(3)	168.	(1)
34.	(3)	79.	(2)	124.	(4)	169.	(3)
35.	(4)	80.	(4)	125.	(1)	170.	(4)
36.	(2)	81.	(2)	126.	(1)	171.	(3)
37.	(1)	82.	(2)	127.	(2)	172.	(3)
38.	(2)	83.	(3)	128.	(3)	173.	(2)
39.	(4)	84.	(2)	129.	(4)	174.	(3)
40.	(1)	85.	(3)	130.	(2)	175.	(1)
41.	(1)	86.	(3)	131.	(2)	176.	(4)
42.	(2)	87.	(2)	132.	(4)	177.	(2)
43.	(1)	88.	(2)	133.	(3)	178.	(1)
44.	(1)	89.	(1)	134.	(4)	179.	(1)
45.	(1)	90.	(1)	135.	(4)	180.	(4)

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Hints & Solutions

1. (2)

Sphere 1 (Point P is outside):

$$E_1 = \frac{kQ}{R^2}$$

Sphere 2 (Point P is on surface):

$$E_2 = \frac{k(2Q)}{R^2} = 2\left(\frac{kQ}{R^2}\right)$$

since $E_2 = 2E_1 \Rightarrow E_2 > E_1$

(NCERT Ed. 2025-26, 12th, Page No. 35)

2. (1)

By symmetry, forces from opposing pairs cancel. The net force is due to the single "missing" charge at the 6th vertex effectively pulling the center.

$$F = \frac{kQq}{r^2} = \frac{9\times10^9\times10\times10^{-6}\times1\times10^{-6}}{1^2}$$

$$= 90\times10^{-3} = 0.09 \text{ N}$$

(NCERT Ed. 2025-26, 12th, Page No. 12)

3. (4)

Wave speed $v = \frac{\omega}{k} = \frac{400}{20} = 20$ m/s

$\mu = 40$ g/m $= 0.04$ kg/m

Tension $T = \mu v^2 = 0.04\times(20)^2$

$= 0.04\times400 = 16$ N

(NCERT Ed. 2025-26, 11th, Page No. 285)

4. (1)

Stress at the ceiling is maximum

$$\frac{(10+m)g}{A} \le 2\times10^6$$

$$\frac{(10+dAL)g}{A} \le 2\times10^6$$

$$\Rightarrow (10+dAL)g \le 6\times10^2$$

$$L_{\max} = \frac{50}{6} \text{ m} = 8.33 \text{ m}$$

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5. (2)

At the time of maximum speed of point P, the rod should be vertical, $\Delta K + \Delta U = 0$

$$\left[\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 - 0\right] + \left[-mg\frac{l}{2}\right] = 0$$

$$\omega = \sqrt{\frac{3g}{l}}$$

$$v_p = \omega l = \sqrt{3gl}$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

6. (4)

For maximum force between two parts of a fixed total charge Q, the charge must be divided equally $\left(q = \frac{Q}{2}\right)$. $q = \frac{10\mu C}{2} = 5\mu C$.

Thus, $5\mu C$ and $5\mu C$.

(NCERT Ed. 2025-26, 12th, Page No. 7)

7. (4)

Time from 0 to $\frac{A}{2}$ (t_1):

In SHM, the displacement equation starting from the mean position is $x = A\sin(\omega t)$.

For $x = \frac{A}{2}$:

$$\frac{A}{2} = A\sin(\omega t_1) \Rightarrow \sin(\omega t_1) = \frac{1}{2} \Rightarrow \omega t_1 = \frac{\pi}{6}.$$

Since $\omega = \frac{2\pi}{T}$, this corresponds to time $t_1 = \frac{T}{12}$

Time from 0 to A (Total time to extreme):

To reach amplitude A, the phase is $\frac{\pi}{2}$.

This corresponds to time $\frac{T}{4}$.

Time from $\frac{A}{2}$ to A (t_2):

$t_2 = (\text{Time to } A) - (\text{Time to } \frac{A}{2})$

$$t_2 = \frac{T}{4} - \frac{T}{12} = \frac{3T}{12} - \frac{T}{12} = \frac{2T}{12} = \frac{T}{6}$$

Comparing the two times: $t_2(\frac{T}{6})$ is exactly twice as long as $t_1(\frac{T}{12})$.

Given $t_1 = 2$ seconds.

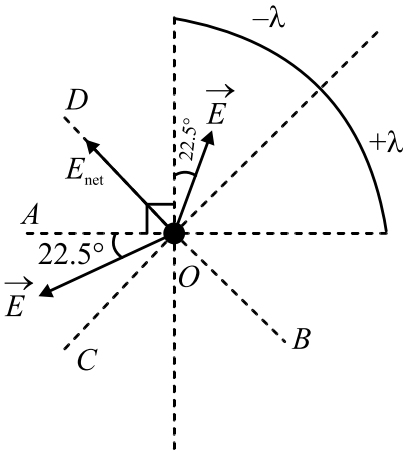
Therefore, $t_2 = 2\times2 = 4$ seconds.

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8. (4)

The electric field due to a circular arc is along its bisector.



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9. (1)

From equation, $k = 5\pi \Rightarrow \frac{2\pi}{\lambda} = 5\pi \Rightarrow \lambda = 0.4$ m

For 3rd harmonic (both ends fixed), $L = \frac{3\lambda}{2}$

$L = \frac{3(0.4)}{2} = 0.6$ m

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10. (1)

Position vector $\vec{r} = \hat{i} + \sqrt{3}\hat{j}$

Magnitude $r = \sqrt{1^2 + (\sqrt{3})^2} = 2$ m

$\vec{E} = \frac{kq}{r^3}\vec{r} = \frac{9\times10^9 \times (1/9)\times10^{-6}}{2^3}\left(\hat{i} + \sqrt{3}\hat{j}\right)$

$= \frac{1000}{8}\left(\hat{i} + \sqrt{3}\hat{j}\right) = 125\left(\hat{i} + \sqrt{3}\hat{j}\right)$

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11. (4)

In cyclic process, $\Delta U = 0$ only

$\Delta Q = \Delta U + \Delta W$

$\Rightarrow \Delta Q = \Delta W \neq 0$

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12. (2)

Maximum velocity depends on amplitude and spring constant as $v_{max} \propto A\sqrt{k}$.

Since v_{max} is equal for both, amplitude is inversely proportional to the square root of k:

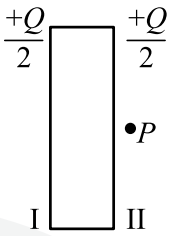
$A \propto \frac{1}{\sqrt{k}}$

Ratio $\frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}} = \sqrt{\frac{400}{100}} = 2:1$

(NCERT Ed. 2025-26, 11th, Page No. 267)

13. (2)

Redistribution of charge on both faces.



At point *P*, electric field due to both faces will add because they are in same direction

$E_{net} = E_I + E_{II} = 2E_{II}$

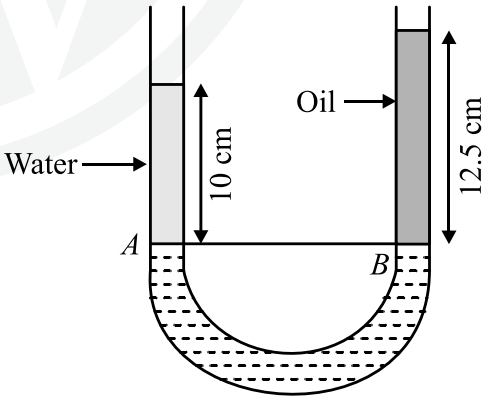
E due to each face is given by $\frac{\sigma}{2\varepsilon_0}$

$E_{each} = \frac{Q}{2A(2\varepsilon_0)} = \frac{Q}{4A\varepsilon_0}$

$E_{net} = \frac{Q}{2A\varepsilon_0}$

(NCERT Ed. 2025-26, 12th, Page No. 63)

14. (1)



$P_A = P_B$

$P_0 + (\rho_w g \times h_{water}) = P_0 + (\rho_{oil} g \times h_{oil})$

$\rho_w \times h_{water} = \rho_{oil} \times h_{oil}$

$\Rightarrow \frac{\rho_{oil}}{\rho_w} = \frac{h_{water}}{h_{oil}}$

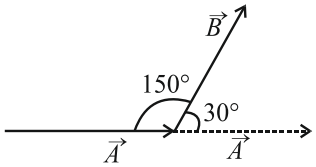
$= \frac{10}{12.5} = 0.8$

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15. (2)

Angle between two vectors are measured either from head to head or tail to tail.



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16. (2)

$$\lambda = \frac{375}{500} = \frac{3}{4}$$

$$\Delta\phi = \frac{2\pi}{\lambda} . \Delta x$$

$$\Rightarrow \frac{\pi}{3} = \frac{2\pi}{\frac{3}{4}} \times (\Delta x)$$

$$\Rightarrow \Delta x = \frac{1}{8} = 0.125\,\text{m}$$

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17. (1)

$$\phi = ES\cos\theta = 3\times10^3 \times (10\times10^{-2})^2 \times \cos 60°$$

$$= 3\times10^3 \times 10^{-2} \times \frac{1}{2} = 15\ \text{Nm}^2/\text{C}.$$

(NCERT Ed. 2025-26, 12th, Page No. 22)

18. (1)

Magnitude of electric field increases as we move near to the source charge. Force on charges will be different in non-uniform field.

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19. (1)

A wave is travelling in negative x direction is represented by $y = A\sin\frac{2\pi}{\lambda}(vt + x)$

$$y = 0.1\sin\left(\frac{2\pi}{60}\right)(360t + x)\,\text{m}$$

$$y = 0.1\sin\left(12\pi t + \frac{2\pi}{60}x\right)\text{m}$$

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20. (2)

$$\sin 30° = \frac{h}{40}$$

$$h = 40\times\frac{1}{2} = 20\,\text{cm}$$

Position of centre of mass $= \frac{h}{4} = \frac{20}{4} = 5\,\text{cm}.$

(NCERT Ed. 2025-26, 11th, Page No. 97)

21. (2)

Electric field inside a hollow or solid conductor under electrostatic condition is zero.

(NCERT Ed. 2025-26, 12th, Page No. 62)

22. (4)

Springs having spring constant $2k$ and $2k$ are in series.

$$\frac{1}{k'} = \frac{1}{2k} + \frac{1}{2k}$$

$$k' = k$$

$$k_{eq} = k + k = 2k$$

$$f = \frac{1}{2\pi}\sqrt{\frac{2k}{M}}$$

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23. (3)

At $t = \frac{3T}{4}$, particle is at mean position

$\Rightarrow$ Force = 0

At $t = T$, particle is at extreme position

$\Rightarrow$ Acceleration is maximum

At $t = \frac{T}{4}$, particle is at mean position

$\Rightarrow$ Speed is maximum

At $t = \frac{T}{2}$, particle is at extreme position

$\Rightarrow$ Speed = 0 $\Rightarrow$ K.E. = 0

$\Rightarrow$ P.E. is maximum.

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24. (4)

$$\frac{d^2x}{dt^2} + 32x = 0$$

On comparing with standard equation

$$\omega = \sqrt{32} = \frac{2\pi}{T}$$

$$\text{T} = \frac{2\pi}{4\sqrt{2}} = \frac{\pi}{2\sqrt{2}}\,\text{s}$$

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25. (4)

Change in angular momentum = Angular impulse

[Angular momentum] = [Angular impulse]

$\Rightarrow [mvr] = [ML^2T^{-1}]$

(NCERT Ed. 2025-26, 11th, Page No. 124)

26. (4)

$(K.E)_T = \frac{3K_BT}{2}$

$(K.E)_R = K_BT$

$\frac{(K.E)_T}{(K.E)_R} = \frac{3}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 253)

27. (1)

Frictional force will only act on the lower surface.

$12 - 0.1 \times 50 = 5a$

$\Rightarrow a = \frac{7}{5}\ m/s^2$

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28. (2)

$\frac{a_1}{a_2} = \frac{3}{4}$

$\frac{I_{max}}{I_{min}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{(3+4)^2}{(3-4)^2} = \frac{49}{1}$

(NCERT Ed. 2025-26, 12th, Page No. 264)

29. (1)

$v_{particle}^{max} = A\omega$ and $v_{wave} = \frac{\omega}{k}$.

Given $A\omega = 3\left(\frac{\omega}{k}\right) \Rightarrow k = \frac{3}{A}$.

Since $k = \frac{2\pi}{\lambda}$, we get $\frac{2\pi}{\lambda} = \frac{3}{A} \Rightarrow \lambda = \frac{2\pi A}{3}$.

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30. (4)

Length of string $L = \frac{\text{Mass}}{\text{density}} = \frac{0.5}{0.1} = 5\ m$.

For fundamental mode,

$v = 2Lf = 2(5)(100) = 1000\ m/s$

(NCERT Ed. 2025-26, 11th, Page No. 291)

31. (2)

$V_P = -V_W \times$ slope

V_W is negative

At *A, B, H* slope is positive, therefore particle velocity is upwards

At *D, E, F* slope is negative, therefore particle velocity is downward.

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32. (3)

$x = 6\sin 4t + 8\cos 4t$

$x = \sqrt{(6)^2 + (8)^2}\sin(4t + \phi)$

$x = 10\sin(4t + \phi)$

$v = \frac{dx}{dt} = 40\cos(4t + \phi)$

$V_{max} = 40$ units

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33. (2)

$\sqrt{x} = t - 3 \Rightarrow x = (t-3)^2$

$\Rightarrow$ speed $V = 2(t-3)$ [differentiating w.r.t. t]

if $t = 0, v = -6$

if $t = 6, v = +6$

using work energy theorem

$\therefore W = \Delta K = K_f - K_i = 0$

(NCERT Ed. 2025-26, 11th, Page No. 73)

34. (3)

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{10}{0.1}} = 10\ m/s$

$t = \frac{0.50}{10} = 0.05 s$

(NCERT Ed. 2025-26, 11th, Page No. 285)

35. (4)

For constraint relation

$a_{10}\cos 37° = a_2$

$\Rightarrow a_{10} \times \frac{4}{5} = 5$

$\Rightarrow a_{10} = \frac{25}{4}\ m/s^2$

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36. (2)

Since, the negative charge of dipole is near to the line charge as compared to the positive charge of the dipole. Hence, the dipole is repelled away from the line charge.

(NCERT Ed. 2025-26, 12th, Page No. 27)

37. (1)

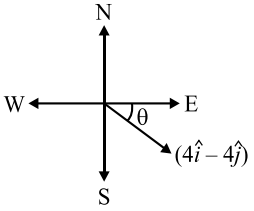
The gravitational field is towards the center of the sphere.

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38. (2)

Displacement $= \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC}$

$= 8\hat{i} + (-4\hat{j}) + (-4\hat{i}) = 4\hat{i} - 4\hat{j}$



| displacement | $= \sqrt{(4)^2 + (4)^2}$

$= 4\sqrt{2}$ km

So, Displacement

$= 4\sqrt{2}$ km, 45° S of E

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39. (4)

Length of line inside sphere $L_s = 2a$

Length inside cube (body diagonal) $L_c = a\sqrt{3}$

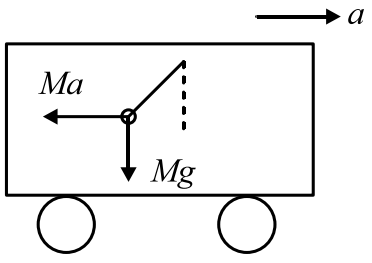
Flux $\phi = \dfrac{\lambda L}{\epsilon_0}$

Ratio $\dfrac{\phi_c}{\phi_s} = \dfrac{L_c}{L_s} = \dfrac{a\sqrt{3}}{2a} = \dfrac{\sqrt{3}}{2}$

$\phi_c = \dfrac{\sqrt{3}}{2}\phi_0$

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40. (1)



$g_{eff} = \sqrt{a^2 + g^2}$

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41. (1)

For small oscillations, use approximation

$1 - \cos\theta \approx \dfrac{\theta^2}{2}$: $U(x) = 10\left[\dfrac{(2x)^2}{2}\right] = 20x^2$

Compare with standard SHM energy $U = \dfrac{1}{2}kx^2$:

$\dfrac{k}{2} = 20 \Rightarrow k = 40$ N/m

$T = 2\pi\sqrt{\dfrac{m}{k}} = 2\pi\sqrt{\dfrac{2}{40}} = 2\pi\sqrt{\dfrac{1}{20}} = \dfrac{\pi}{\sqrt{5}}s$

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42. (2)

According to Gauss theorem $\phi_{net} = \dfrac{q_{enclosed}}{\varepsilon_0}$

(NCERT Ed. 2025-26, 11th, Page No. 267)

43. (1)

Total Energy $E = K + U = 60 + 40 = 100$ J.

Energy formula: $E = \dfrac{1}{2}m\omega^2 A^2$.

$100 = \dfrac{1}{2}(2)(10)^2 A^2 \Rightarrow 100 = 100A^2 \Rightarrow A = 1$ m

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44. (1)

Using phasors: Components along *x*-axis

$= A + A\cos(60^\circ) + A\cos(-60^\circ)$

$= A + 0.5A + 0.5A = 2A$

Y-components cancel out. Resultant Amplitude $A_{res} = 2A$.

Since $I \propto A^2$, $I_{res} = (2)^2 I_0 = 4I_0$.

(NCERT Ed. 2025-26, 12th, Page No. 264)

45. (1)

Formula for line charge: $E = \dfrac{2k\lambda}{r}$

$\lambda = \dfrac{E \cdot r}{2k} = \dfrac{1.8\times10^5 \times 0.1}{2\times9\times10^9} = \dfrac{1.8\times10^4}{18\times10^9}$

$= 10^{-6}$ C/m $= 1\mu$C/m

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46. (4)

Metamers are the isomers having same molecular formula and same functional group but different alkyl or aryl groups on either side of functional group.

(NCERT Ed. 2025-26 11th Part-II Page No. 271)

47. (4)

Carbonyl compounds having α-H atom show tautomerism.

(NCERT Ed. 2025-26 11th Part-II Page No. 317)

48. (4)

In carbanion, hyperconjugation does not take place.

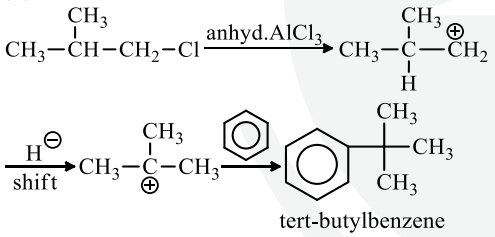
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49. (3)

-  : $2\pi e^-$, planar (aromatic)
-  : $6\pi e^-$, planar (aromatic)
-  : $4\pi e^-$, planar (anti-aromatic)
-  : $2\pi e^-$, planar (aromatic)

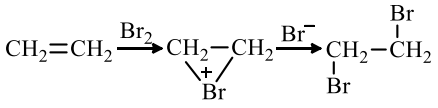
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50. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 322)

51. (1)



(NCERT Ed. 2025-26 11th Part-II Page No. 311)

52. (2)

$Ca(H_2PO_3)_2 : +2+2(2+x-6)=0$

$x=+3$

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

53. (4)

- In vacuum all types of electromagnetic radiations, regardless of wavelength, travel at same speed, i.e., speed of light.
- Black body radiation phenomenon support the particle nature of electromagnetic radiation.

(NCERT Ed. 2025-26 11th Part-I Page No. 38)

54. (2)

- 5.6 L: $\frac{5.6}{22.4}\times 28 = 7$ g
- 3.022×10^{22} molecule: $\frac{3.022\times10^{22}}{6.022\times10^{23}}\times 28 = 1.4$ g
- 0.5 g molecule: $0.5 \times 28 = 14$ g
- 0.01 mole: $0.01 \times 28 = 0.28$ g

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

55. (4)

- Molecules in gaseous phase have greater freedom of movement. i.e. have greater randomness than those in solid phase, hence entropy increases.
- On increasing temperature of a solid, entropy increases.
- As number of gaseous moles increases, randomness increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 159)

56. (2)

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

eq (1)	p	p
eq (2)	2p	p′

From eq (1), $K_p = (p)(p) = p^2$

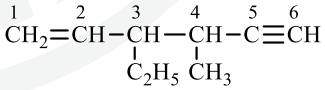
total pressure = p + p = 2p

From eq (2), $K_p = (2p)(p')$

$$p^2 = 2p \times p'; \quad p' = \frac{p}{2} \quad \therefore \frac{p'}{p} = \frac{1}{2}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 180)

57. (1)



3-Ethyl-4-methylhex-1-en-5-yne

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

58. (3)

$\Delta T_f = i \times K_f \times \text{molality}$

$$1.024 = \frac{i \times 5.12 \times \left(\frac{2.44}{122}\right)}{50\times10^{-3}}$$

$i = 0.5$

$$2C_6H_5COOH \xrightarrow{C_6H_6} (C_6H_5COOH)_2$$

(*i*)	1	0
(f)	$1-\alpha$	$\frac{\alpha}{2}$

$$i = \frac{1-\alpha+\frac{\alpha}{2}}{1} = 1-\frac{\alpha}{2}; \quad 0.5 = 1-\frac{\alpha}{2} \quad \therefore \alpha = 1$$

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[Contd...

59. (3)

Anode: $Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$

On account of overpotential of oxygen, $Cl^-(aq)$ is oxidized to $Cl_2(g)$

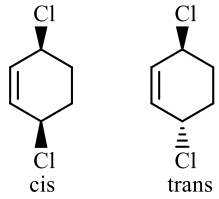
(NCERT Ed. 2025-26 12th Part-I Page No. 53)

60. (3)

Aliphatic alcohols and phenols are different family/functional groups.

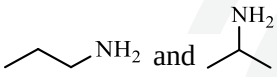
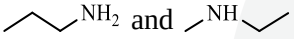
(NCERT Ed. 2025-26 11th Part-II Page No. 270)

61. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 308)

62. (4)

-  : position isomers
-  : functional isomers

(NCERT Ed. 2025-26 11th Part-II Page No. 270)

63. (1)

Compound	Acidic H-atom
$H-C\equiv C-H$	2
$CH_3-C\equiv C-H$	1
$CH_3CH_2-C\equiv C-H$	1
$CH_3-C\equiv C-CH_3$	0

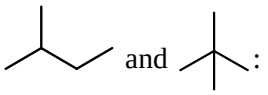
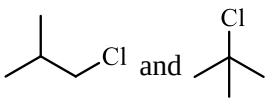
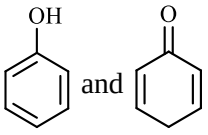
(NCERT Ed. 2025-26 11th Part-II Page No. 316)

64. (3)

Dehydrohalogenation in presence of bulky base is governed by Hoffmann rule and less stable alkene is major product formed by E_2 pathway.

(NCERT Ed. 2025-26 11th Part-II Page No. 316)

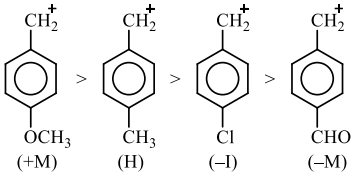
65. (2)

-  : Chain isomers
-  : Position isomers
-  : Tautomers
- CH_3CH_2OH and CH_3OCH_3: Functional isomers

(NCERT Ed. 2025-26 11th Part-II Page No. 270)

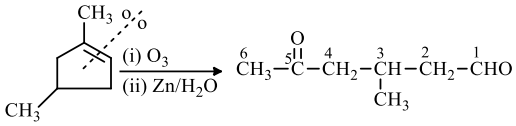
66. (1)

Stability order:



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67. (3)



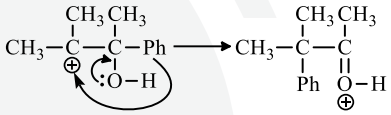
(NCERT Ed. 2025-26 11th Part-II Page No. 313)

68. (3)

Cyclopentene (C_5H_8) and  (C_5H_{10}) have different molecular formula.

(NCERT Ed. 2025-26 11th Part-II Page No. 270)

69. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 272)

70. (1)

- $CH_4 + I_2 \rightleftharpoons CH_3I + HI$

  $HI \xrightarrow[OA]{HNO_3} I_2$
- In nitration of benzene, HNO_3 acts as a base and H_2SO_4 acts as an acid.

(NCERT Ed. 2025-26 11th Part-II Page No. 302)

71. (4)

Chlorine undergo disproportionation in the alkaline medium.

$Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + H_2O(\ell)$

(NCERT Ed. 2025-26 11th Part-II Page No. 244)

72. (1)

$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$; $\frac{1}{\lambda} = R \times 1^2 \times \left(\frac{1}{3^2} - \frac{1}{4^2}\right)$

$\frac{1}{\lambda} = \frac{7R}{144}$ $\quad \therefore \lambda = \frac{144}{7R}$

(NCERT Ed. 2025-26 11th Part-I Page No. 45)

73. (2)

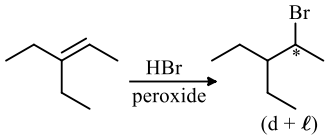
$\underset{4\,g}{2\,mol\ H_2} + \underset{32\,g}{1\,mol\ O_2} \rightarrow 2H_2O$

$\therefore$ for 3 g H_2 only 24 g O_2 is required

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

[Contd...

74. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 312)

75. (4)

$[CH_3COO^-] = 0.1 \times 2 = 2 \times 10^{-1}\,M$

For anionic hydrolysis:

$$pH = \frac{1}{2}(pK_w + pK_a + \log C)$$

$$= \frac{1}{2}\left[14 + 4.74 + \log(2 \times 10^{-1})\right]$$

$$= \frac{1}{2}[18.74 - 1 + 0.3010] = 9.02$$

(NCERT Ed. 2025-26 11th Part-I Page No. 202)

76. (3)

A mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law because chloroform molecule is able to form hydrogen bond with acetone molecule.

(NCERT Ed. 2025-26 12th Part-I Page No. 14)

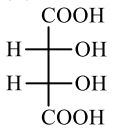
77. (4)

The conductivity of electrolytic solution depends on:

- The concentration of electrolyte
- Nature of solvent
- Temperature
- Size of the ions produced
- Nature of electrolyte added.

(NCERT Ed. 2025-26 12th Part-I Page No. 43)

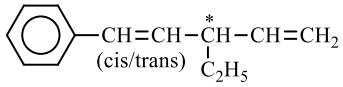
78. (3)



Meso tartaric acid

(NCERT Ed. 2025-26 12th Part-II Page No. 177)

79. (2)



The compound contain two stereogenic sites, so total stereoisomers = $2^2 = 4$

(NCERT Ed. 2025-26 12th Part-II Page No. 176)

80. (4)

All carbon are sp^2 hybridised as negative charge is involved in resonance.

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

81. (2)

More the double bond character between C and Cl due to resonance, more will be C-Cl bond energy.

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

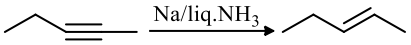
82. (2)

Benzenehexachloride, $C_6H_6Cl_6$ is also called gammaxane.

(NCERT Ed. 2025-26 11th Part-II Page No. 324)

83. (3)

Trans alkene will obtain



(NCERT Ed. 2025-26 11th Part-II Page No. 309)

84. (2)

Enantiomers have same physical properties like melting point and boiling point.

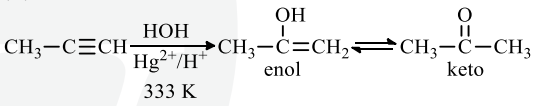
(NCERT Ed. 2025-26 12th Part-II Page No. 177)

85. (3)

–I effect: $-\overset{+}{N}H_3 > -CN > -F$

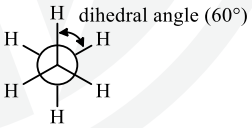
(NCERT Ed. 2025-26 11th Part-II Page No. 274)

86. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 317)

87. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 305)

88. (2)

- Electrophile: $AlCl_3, Co^{3+}, SnCl_4, BF_3$
- Nucleophile: H_2O, NH_3, C_2H_4

(NCERT Ed. 2025-26 11th Part-II Page No. 273)

89. (1)

- As the branching increases, boiling point of alkane decreases.
- As the molar mass increases, boiling point of alkane increases.

(NCERT Ed. 2025-26 11th Part-II Page No. 302)

90. (1)

Stability order:

$$\overset{\ominus}{C}H_3 > CH_3-\overset{\ominus}{C}H_2 > (CH_3)_2\overset{\ominus}{C}H > (CH_3)_3\overset{\ominus}{C}$$

(NCERT Ed. 2025-26 11th Part-II Page No. 274)

[Contd...

91. (2)
Mayr was awarded the three prizes widely regarded as the triple crown of biology: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999.
(NCERT Ed. 2025-26, 11th, Page 2)

92. (1)
Parenchyma forms the major component within organs. The cells of the parenchyma are generally isodiametric. They may be spherical, oval, round, polygonal or elongated in shape. Their walls are thin and made up of cellulose. They may either be closely packed or have small intercellular spaces. The parenchyma performs various functions like photosynthesis, storage, secretion.
(NCERT Ed. 2022-23, 11th, Page 86)

93. (1)
Both apical meristems and intercalary meristems are primary meristems because they appear early in life of a plant and contribute to the formation of the primary plant body.
(NCERT Ed. 2022-23, 11th, Page 84-85)

94. (2)
The taxonomic categories from species to kingdom in ascending order are as follows:
Species → Genus → Family → Order → Class → Phylum → Kingdom
(NCERT Ed. 2025-26, 11th, Page 8)

95. (1)
A mature sieve element possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus. The functions of sieve tubes are controlled by the nucleus of companion cells.
(NCERT Ed. 2022-23, 11th, Page 88)

96. (4)
Tracheids are elongated or tube like cells with thick and lignified walls and tapering ends. Vessel is a long cylindrical tube-like structure made up of many cells called vessel members. Xylem parenchyma cells are living and thin-walled, and their cell walls are made up of cellulose. Xylem fibres have highly thickened walls and obliterated central lumens.
(NCERT Ed. 2022-23, 11th, Page 87)

97. (2)
In a root tip, the zone, proximal to region of elongation, is called the region of maturation.
(NCERT Ed. 2025-26, 11th, Page 59)

98. (1)
Sclerenchyma cells (fibres and sclereids) are dead at maturity and have thick, lignified cell walls. They lack protoplasm and do not contain chloroplasts.
(NCERT Ed. 2025-26, 11th, Page 72)

99. (3)
Within angiosperms, the monocots and dicots are also seen to be anatomically different. Internal structures also show adaptations to diverse environments.
(NCERT Ed. 2025-26, 11th, Page 71)

100. (3)
Katherine Esau reported in her early publications that the curly top virus spreads through a plant via the food-conducting or phloem tissue.
(NCERT Ed. 2025-26, 11th, Page 56)

101. (1)
R.H. Whittaker (1969) proposed a Five Kingdom Classification. The kingdoms defined by him were named Monera, Protista, Fungi, Plantae and Animalia. The main criteria for classification used by him include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships. All single-celled eukaryotes are placed under Protista. Monera and Protista have a cellular level of organization.
Fungi have a multicellular/loose tissue level.
Plantae have tissue/organ level organization.
Animalia is the only kingdom where the organ system level of body organization is widely prevalent.
(NCERT Ed. 2025-26, 11th, Page 19)

102. (4)
The conducting tissue system (vascular system) contains both living and dead cells. In xylem, while tracheids, vessels, and fibers are dead, the xylem parenchyma is a living cell. In phloem, the sieve tube elements, companion cells, and phloem parenchyma are living cells; only the phloem fibers are dead.
(NCERT Ed. 2025-26, 11th, Page 73)

103. (1)
When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, the arrangement is called radial. In conjoint type of vascular bundles, the xylem and phloem are jointly situated along the same radius. In dicotyledonous stems, cambium is present between phloem and xylem, hence the vascular bundles are called open vascular bundles. In the monocotyledons, the vascular bundles have no cambium present in them, hence referred to as closed.
(NCERT Ed. 2025-26, 11th, Page 73)

[Contd...

104. (2)

Scientific names do not include the local names of organisms. The first component of a scientific name is the genus, and according to the rules, it must always start with a capital letter. It is followed by the specific epithet. The author's name (in abbreviated form) is written at the end of the scientific name, after the specific epithet.

(NCERT Ed. 2025-26, 11th, Page 5)

105. (4)

Casparian strips are present in the roots, not in stems.

(NCERT Ed. 2025-26, 11th, Page 75)

106. (1)

Placentation	Example
Marginal	Pea
Axile	Lemon
Parietal	*Argemone*
Free central	*Primrose*

(NCERT Ed. 2025-26, 11th, Page 65)

107. (2)

In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells.

(NCERT Ed. 2025-26, 11th, Page 77)

108. (3)

The epidermis is the outermost protective layer of the stem, consists of a few layers of collenchymatous cells. Covered with a thin layer of cuticle, it may bear trichomes and a few stomata. The innermost layer of the cortex is called the endodermis (single layer of cells).

(NCERT Ed. 2025-26, 11th, Page 75)

109. (2)

When the bulliform cells are flaccid due to water stress, they make the grass leaves curl inwards to minimise water loss.

(NCERT Ed. 2025-26, 11th, Page 77)

110. (2)

All three (human, dogs and cats) are in the Kingdom Animalia, Phylum Chordata, and Class Mammalia. The classification diverges at the Order level, where humans belong to the Order Primates, while dogs and cats belong to the Order Carnivora. Therefore, the lowest, or most specific, taxonomic category they all share is Class Mammalia.

(NCERT Ed. 2025-26, 11th, Page 7-8)

111. (1)

Phloem parenchyma is absent in most of the monocotyledons. Phloem fibres (bast fibres) are made up of sclerenchymatous cells.

(NCERT Ed. 2022-23, 11th, Page 88)

112. (2)

In a dorsiventral leaf, abaxial epidermis generally bears more stomata than the adaxial epidermis.

(NCERT Ed. 2025-26, 11th, Page 76)

113. (4)

Sclerenchyma consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits. They are usually dead and without protoplasts. On the basis of variation in form, structure, origin and development, sclerenchyma may be either fibres or sclereids. Sclerenchyma provides mechanical support to organs. The monocot stem has a sclerenchymatous hypodermis.

(NCERT Ed. 2022-23, 11th, Page 86)

114. (3)

Diatoms, *Cycas* and *Gonyaulax* are photosynthetic autotrophs. *Agaricus* is a saprotroph.

(NCERT Ed. 2025-26, 11th, Page 14-18)

115. (1)

In a dicot root, all tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

(NCERT Ed. 2025-26, 11th, Page 74)

116. (4)

Plants that undergo secondary growth usually possess both a vascular cambium and a cork cambium. Having only one cotyledon defines a monocot. Parallel venation is a characteristic feature of monocots. Monocots generally lack secondary growth. An "open" vascular bundle is one that contains a layer of cambium between the xylem and phloem. This cambium is what allows the plant to produce secondary vascular tissues.

(NCERT Ed. 2025-26, 11th, Page 73)

117. (2)

Ulothrix and *Spirogyra* show isogamy. *Volvox* and *Fucus* show oogamy.

(NCERT Ed. 2025-26, 11th, Page 24)

118. (3)

In grasses, the guard cells are dumb-bell shaped.

(NCERT Ed. 2025-26, 11th, Page 72)

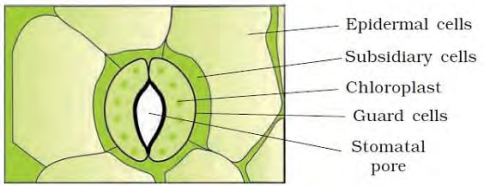
119. (4)

Mustard belongs to family Cruciferae.

(NCERT Ed. 2025-26, 11th, Page 68-69)

[Contd...

120. (3)
The stomatal aperture, guard cells and the surrounding subsidiary cells are together called stomatal apparatus.



(NCERT Ed. 2025-26, 11th, Page 72)

121. (1)
Strobili are present in *Equisetum*. Heterospory is seen in *Selaginella*.

(NCERT Ed. 2025-26, 11th, Page 30)

122. (3)
The conjoint vascular bundles usually have the phloem located only on the outer side of xylem.

(NCERT Ed. 2025-26, 11th, Page 73)

123. (3)

Steroid hormone synthesis	Smooth endoplasmic reticulum
Protein synthesis	Polysomes
Glycoprotein synthesis	Golgi complex
ATP synthesis	Mitochondria

(NCERT Ed. 2025-26, 11th, Page 91, 95-97)

124. (4)
Permanent tissues having all cells similar in structure and function are called simple tissues. The various simple tissues in plants are parenchyma, collenchyma and sclerenchyma. Permanent tissues having many different types of cells are called complex tissues. Xylem and phloem constitute the complex tissues in plants.

(NCERT Ed. 2022-23, 11th, Page 86-87)

125. (1)
The diagram given in the question is a T.S. of a monocot (isobilateral) leaf. It consists of near similar sizes of vascular bundles (except in main veins). The veinlets form a network in dicot (dorsiventral) leaves.

(NCERT Ed. 2025-26, 11th, Page 60)

126. (1)
Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of lateral meristems. Intercalary meristems are primary meristems.

(NCERT Ed. 2022-23, 11th, Page 85)

127. (2)
Mitosis is known as equational division because the number of chromosomes in the daughter cells remains the same as that of the parent cell. Hence, the daughter cells as per the question will have 20 chromosomes each.

(NCERT Ed. 2025-26, 11th, Page 121)

128. (3)
In dicot roots, the tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.

(NCERT Ed. 2025-26, 11th, Page 74)

129. (4)
In dicot roots, the parenchymatous cells which lie between the xylem and the phloem are called conjuctive tissue.

(NCERT Ed. 2025-26, 11th, Page 74)

130. (2)
On the stem the epidermal hairs are called trichomes. The trichomes in the shoot system are usually multicellular. They may be branched or unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

(NCERT Ed. 2025-26, 11th, Page 72)

131. (2)
In monocot stem, water-containing cavities are present within the vascular bundles.

(NCERT Ed. 2025-26, 11th, Page 76)

132. (4)
In a dicot root, the innermost layer of the cortex is called endodermis. It comprises a single layer of barrel-shaped cells without any intercellular spaces.

(NCERT Ed. 2025-26, 11th, Page 74)

133. (3)
Chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA.

(NCERT Ed. 2025-26, 11th, Page 101)

134. (4)
As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root.

(NCERT Ed. 2025-26, 11th, Page 74)

135. (4)
The growth of multicellular organisms is due to mitosis. Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio. A very significant contribution of mitosis is cell repair. The cells of the upper layer of the epidermis, cells of the lining of the gut, and blood cells are being constantly replaced. Meiosis increases the genetic variability in the population of organisms from one generation to the next. Variations are very important for the process of evolution.

(NCERT Ed. 2025-26, 11th, Page 128)

[Contd...

136. (1)

At puberty, there is a marked increase in GnRH secretion from the hypothalamus, which stimulates the pituitary to release FSH and LH. These hormones initiate spermatogenesis in human males.

(NCERT Ed. 2025-26, 12th, Page No. 31)

137. (1)

Digestion is both extracellular and intracellular in coelenterate, not extracellular only.

Adamsia (sea anemone) is a polyp, not a medusa.

(NCERT Ed. 2025-26, 11th, Page No. 41)

138. (1)

The three layers of the alveolar diffusion membrane are

1. Thin squamous epithelium of the alveoli
2. Endothelium of alveolar capillaries
3. Basement substance between them

The total thickness of the diffusion membrane is extremely thin, much less than 1 millimetre, which facilitates rapid gaseous exchange.

(NCERT Ed. 2025-26, 11th, Page No. 188)

139. (3)

In the given diagram X represents Sertoli cells whereas Y represents interstitial cells. Sertoli cells provide nutrition to the developing male germ cells. Interstitial (Leydig) cells synthesise and secrete testicular hormones called androgens (testosterone).

(NCERT Ed. 2025-26, 12th, Page No. 27, 28)

140. (1)

List-I	List-II
Balanoglossus	Proboscis gland
Pheretima	Nephridia
Taenia	Flame cells
Locusta	Malpighian tubules

(NCERT Ed. 2025-26, 11th, Page No. 42, 43, 44, 45)

141. (3)

In a standard ECG:

- P wave → Depolarisation of atria
- QRS complex → Depolarisation of ventricles
- T wave → Repolarisation of ventricles

(NCERT Ed. 2025-26, 11th, Page No. 201)

142. (3)

The corpus luteum mainly secretes progesterone, which is essential for maintenance of the endometrium, not estrogen in large amounts.

(NCERT Ed. 2025-26, 12th, Page No. 34, 35)

143. (2)

All three animals belong to the same phylum — Mollusca:

- *Dentalium*
- *Pinctada*
- *Chaetopleura*

(NCERT Ed. 2025-26, 11th, Page No. 45)

144. (3)

List-I	List-II
PCT	Reabsorption of 70-80% of electrolytes and water
Descending limb of Henle's Loop	Permeable to water, impermeable to electrolytes
Ascending limb of Henle's Loop	Impermeable to water, permeable to electrolytes
DCT	Conditional reabsorption of Na^+ and water

(NCERT Ed. 2025-26, 11th, Page No. 209)

145. (3)

The myometrium (middle muscular layer of the uterus) shows strong contractions during childbirth, helping in delivery of the baby.

Although the uterus is inverted pear-shaped, it is supported by ligaments, not tendons, attached to the pelvic wall.

(NCERT Ed. 2025-26, 12th, Page No. 29, 30)

146. (2)

Arthropoda is the largest, not the second-largest, phylum of Animalia.

Arthropods have an open circulatory system, not closed.

(NCERT Ed. 2025-26, 11th, Page No. 44)

147. (3)

According to the sliding filament theory during contraction:

- A band remains constant in length
- I band shortens
- H zone disappears or reduces during contraction
- Actin filaments slide towards (into) the A band, not away

(NCERT Ed. 2025-26, 11th, Page No. 222, 223)

148. (3)

Placenta also acts as an endocrine tissue and produces several hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, etc.

In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovary.

(NCERT Ed. 2025-26, 12th, Page No. 37)

[Contd...

149. (2)

The midbrain is located between the thalamus/hypothalamus of the forebrain and the pons of the hindbrain. It is traversed by the cerebral aqueduct, which connects the third and fourth ventricles. Its dorsal part forms the corpora quadrigemina, consisting of four round swellings involved in visual and auditory reflexes.

(NCERT Ed. 2025-26, 11th, Page No. 236)

150. (1)

Sponges (phylum Porifera) possess a well-developed water transport or canal system, which is a defining feature of the group. This system allows continuous flow of water through the body and is essential for food collection, gaseous exchange, and removal of metabolic wastes.

(NCERT Ed. 2025-26, 11th, Page No. 40)

151. (3)

Cortisol is a steroid hormone derived from cholesterol. Steroid hormones are lipid-soluble, diffuse through the cell membrane, bind to intracellular (cytoplasmic or nuclear) receptors, and directly regulate gene expression.

Insulin and Glucagon are peptide hormones; Epinephrine is an amino-acid derivative (act via membrane receptors).

(NCERT Ed. 2025-26, 11th, Page No. 248)

152. (2)

In humans, embryonic development follows this sequence: Zygote → Morula → Blastocyst → Implantation. The zygote undergoes cleavage to form a morula, which then develops into a blastocyst that gets implanted in the uterine wall.

(NCERT Ed. 2025-26, 12th, Page No. 36, 37)

153. (1)

Cyclostomes (e.g., *lampreys*) are ectoparasites on some fishes.

They have an elongated body with 6–15 pairs of gill slits.

Their body is devoid of scales and paired fins.

Cyclostomes are jawless; they have a circular sucking mouth without jaws.

They are marine but migrate to freshwater for spawning.

(NCERT Ed. 2025-26, 11th, Page No. 47)

154. (2)

In the male reproductive system of a cockroach:

- The mushroom glands (accessory glands) are located in the 6th and 7th abdominal segments.
- They function as an accessory reproductive gland.

(NCERT Ed. 2020-21, 11th, Page No. 114)

155. (4)

The foetal ejection reflex (parturition/childbirth) involves:

- A fully developed foetus → triggers uterine stretching
- Placenta → helps release prostaglandins
- Oxytocin from the maternal pituitary → stimulates strong uterine contractions

Prolactin is not involved in the ejection reflex; it mainly stimulates milk production after birth.

(NCERT Ed. 2025-26, 12th, Page No. 38)

156. (3)

The given characteristics describe birds (Class Aves):

- Skin is dry and mostly without glands; some oil glands are present near the tail (uropygial gland).
- Heart is completely four-chambered.
- They are warm-blooded (homoiothermic).
- Air sacs connected to lungs help in efficient respiration.
- Bones are hollow with air cavities (pneumatic bones) to reduce weight for flight.

(NCERT Ed. 2025-26, 11th, Page No. 49, 50)

157. (3)

Vital Capacity (VC) = ERV + TV + IRV.

VC = 1000 + 500 + 2500 = 4000 mL.

(NCERT Ed. 2025-26, 11th, Page No. 187)

158. (4)

Each spermatogonium is diploid, but it contains 46 chromosomes in total, not 46 pairs. Humans have 46 chromosomes = 23 pairs.

Human sex is determined by sex chromosomes:

Female: XX

Male: XY

Egg (from mother): Always contributes X chromosome

Sperm (from father): Can contribute X or Y chromosome

X sperm + X egg → Female (XX)

Y sperm + X egg → Male (XY)

So, the father's sperm determines the sex of the baby, not the mother's egg.

(NCERT Ed. 2025-26, 12th, Page No. 31, 36)

159. (3)

Ornithorhynchus (Platypus) is an egg-laying (oviparous) mammal. The others mammals are viviparous.

(NCERT Ed. 2025-26, 11th, Page No. 51)

160. (2)

Anti-Rh antibodies are administered to the Rh-negative mother, not the child, immediately after the delivery of an Rh-positive baby. Administering it to the child will not prevent *erythroblastosis foetalis* in subsequent pregnancies.

The anti-Rh antibodies in the mother destroy any Rh-positive fetal red blood cells that entered her bloodstream during childbirth, preventing her immune system from being sensitized.

(NCERT Ed. 2025-26, 11th, Page No. 196)

161. (3)

- In oogenesis, the primary oocyte completes the first meiotic division before ovulation, forming a secondary oocyte and the first polar body.
- The secondary oocyte arrests at metaphase II and only completes the second meiotic division upon fertilisation, when the sperm fuses with the oocyte.
- This ensures that the oocyte contributes half of the chromosomes to the zygote.

(NCERT Ed. 2025-26, 12th, Page No. 36)

162. (2)

The notochord in Urochordata is present in the larval tail, and in Cephalochordata, it extends through the body throughout life. The structure present in Urochordata and Cephalochordata is the notochord, not stomochord.

Asterias (starfish) is an echinoderm and lacks a well-developed excretory system. It does not have kidneys or similar organs; excretion occurs by diffusion across the body surface.

(NCERT Ed. 2025-26, 11th, Page No. 45, 46)

163. (2)

Aldosterone is a steroid hormone secreted by the adrenal cortex.

It acts on the distal convoluted tubules and collecting ducts of the nephron to:

- Increase reabsorption of Na^+ (sodium)
- Enhance water reabsorption (follows sodium osmotically)

This increases blood volume and blood pressure.

(NCERT Ed. 2025-26, 11th, Page No. 212)

164. (1)

The pathway of sperm transport in the male reproductive system is as follows:

1. Rete testis → collects sperm from the seminiferous tubules inside the testis.
2. Vasa efferentia → channels sperm from the rete testis to the epididymis.
3. Epididymis → stores sperm and allows them to mature and gain motility.
4. Vas deferens → transports mature sperm from the epididymis to the urethra during ejaculation.

(NCERT Ed. 2025-26, 12th, Page No. 27)

165. (2)

- In chordates, the dorsal nerve cord is a defining feature.

It is:

- Dorsal → lies above the notochord.
- Hollow → forms the central canal in vertebrates.
- The notochord lies ventral to the nerve cord and provides structural support during development.

(NCERT Ed. 2025-26, 11th, Page No. 46)

166. (3)

- The atlanto-axial joint is formed between the atlas (C1) and axis (C2) vertebrae.
- It allows rotation of the head from side to side ("no" movement).
- This joint is a pivot type of synovial joint, where the odontoid process (dens) of the axis acts as the pivot around which the atlas rotates.

(NCERT Ed. 2025-26, 11th, Page No. 227)

167. (4)

The placenta is a temporary organ that connects the embryo/fetus to the maternal uterus via the umbilical cord.

Its functions include:

Exchange of gases – supplies oxygen and removes carbon dioxide.

Nutrition – provides glucose, amino acids, and other nutrients.

Excretion – removes nitrogenous wastes from fetal blood.

Hormone production – secretes hCG, hPL, progesterone, and estrogen.

It does not produce gametes. Haploid gametes are formed by the gonads (testes or ovaries), not the placenta.

(NCERT Ed. 2025-26, 12th, Page No. 37)

168. (1)

Ascaris (a nematode) exhibits sexual dimorphism:

- Males are smaller than females.
- The male has a curved tail, while the female is straight and larger.

The alimentary canal in *Ascaris* is complete: it has a mouth, pharynx, intestine, and anus.

The pharynx is muscular, helping in the ingestion of food.

(NCERT Ed. 2025-26, 11th, Page No. 43)

169. (3)

- Electrical synapses have membranes of the pre- and post-synaptic neurons in very close proximity, often connected by gap junctions, allowing direct flow of ions and electrical current.
- They are faster than chemical synapses because no neurotransmitters are involved.
- They are rare in humans, unlike chemical synapses, which are more common.
- Chemical synapses have a synaptic cleft and rely on neurotransmitter release.

(NCERT Ed. 2025-26, 11th, Page No. 234)

170. (4)

Growing follicles primarily secrete estrogen, not progesterone. Progesterone is mainly secreted by the corpus luteum after ovulation.

(NCERT Ed. 2025-26, 12th, Page No. 34)

171. (3)

List-I	List-II
Addison's disease	Hyposecretion of adrenal cortex hormones
Acromegaly	Hypersecretion of growth hormone in adults
Cretinism	Hypothyroidism during pregnancy
Diabetes mellitus	Deficiency of insulin

(NCERT Ed. 2025-26, 11th, Page No. 241, 242, 244, 246)

172. (3)

Endocrine system: Provides hormonal, slow, and widespread coordination, not rapid point-to-point coordination. That is the function of the nervous system.

Molluscs: Locomotion is mainly by muscular foot, not by well-developed longitudinal and circular muscles like in annelids.

Nereis (aquatic annelid): Has parapodia (lateral appendages) for swimming, not pseudopodia.

Scoliodon (Chondrichthyes): Notochord persists throughout life, forming the main axial support.

(NCERT Ed. 2025-26, 11th, Page No. 43, 44, 47, 239)

173. (2)

The head of sperm contains an elongated haploid nucleus (23 chromosomes).

The acrosome is a cap-like structure over the anterior portion of the head containing enzymes to penetrate the egg.

The middle piece is rich in mitochondria, not ribosomes, to produce ATP for motility.

The tail (flagellum) facilitates sperm motility, essential for fertilization.

(NCERT Ed. 2025-26, 12th, Page No. 32)

174. (3)

Frogs have a three-chambered heart consisting of two atria (left and right) and one ventricle.

The right atrium receives deoxygenated blood from the body, and the left atrium receives oxygenated blood from the lungs and skin.

Both types of blood enter the single ventricle, where partial mixing occurs before being pumped to the body and lungs.

(NCERT Ed. 2025-26, 11th, Page No. 82)

175. (1)

List-I	List-II
Petromyzon	Lamprey
Branchiostoma	Lancelet
Clarias	Magur
Chelone	Turtle

(NCERT Ed. 2025-26, 11th, Page No. 46, 47, 48, 49)

176. (4)

- Alveolar $pO_2 \approx 104$ mmHg
- Oxygenated blood leaving lungs $pO_2 \approx 95$ mmHg

Oxygen diffuses from alveoli (higher pO_2) into blood (lower pO_2). So alveolar pO_2 is slightly higher than in oxygenated blood due to near-equilibrium conditions.

This difference drives oxygen diffusion into the blood.

(NCERT Ed. 2025-26, 11th, Page No. 187)

[Contd...

177. (2)

Fertilisation in humans occurs in the ampullary region of the fallopian tube.

- Sperm must reach the ampulla
- Ovum must reach the ampulla
- Sperm and ovum must meet simultaneously
- Cervix being blocked by a mucus plug is not required; in fact, sperm need to pass through the cervical mucus to reach the uterus and fallopian tube.

(NCERT Ed. 2025-26, 12th, Page No. 35)

178. (1)

Ammonia is highly toxic → requires large amounts of water for excretion → typical of aquatic animals.
Uric acid is least toxic → excreted with minimal water → typical of birds and reptiles.
Bioluminescence is the emission of light by living organisms.
It is well-marked in Cnidarians (like *Obelia*) and some deep-sea fishes, not in Ctenophores.
Ctenophores show iridescence, which is light reflection, not true bioluminescence.

(NCERT Ed. 2025-26, 11th, Page No. 42, 205)

179. (1)

The SAN sets the pace because it has the highest auto-excitability rate among nodal tissues, driving the heart rate.

(NCERT Ed. 2025-26, 11th, Page No. 199)

180. (4)

Seminal plasma is the fluid part of semen secreted mainly by the seminal vesicles, prostate, and bulbourethral glands.
It is rich in:

- Fructose → energy source for sperm motility
- Calcium → important for sperm motility and acrosome reaction
- Enzymes → e.g., proteolytic enzymes, phosphatases, fibrinolysin to aid sperm function and semen liquefaction

(NCERT Ed. 2025-26, 12th, Page No. 28)

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